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Financial Analysis of Intertek Group Plc

Research Project 2

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Abstract

This paper aims to provide an analysis of Intertek Group Plc. It discusses the company's financial position and performance on the basis of financial policies, final account statements and stock market performance. A variety of methods are employed to determine the valuation of the company by taking into account the relevant risk of the market. Various ratios of the business which assess the profitability, liquidity, asset management, efficiency and effectiveness of the company are calculated. These figures are analysed in detail by comparing them with the peer companies and the industry. In order to help explain the undervaluation or overvaluation of the company and providing information to investors about the company's ability to generate profit per share, Price to Equity and Earning Per Share ratios are computed. For complete and accurate evaluation of the company's past financial efficiency, its stock price performance has been taken into account by using the Abnormal Buy and Hold Return strategy. The literature review section also provides the definition of the capital structure of the company and expenditure policy. It comments on dividend per share, the pay-out ratio and dividend yield which allows the dividend policy of the company to be assessed and evaluated. Beta/ Risk of the company is measured to enable a discussion on the previous and prevailing market risk. Since the data is limited to one company, the introduction section provides a brief review of the market and the company. This research concludes by helping investors decide whether to increase or decrease their holdings of the company's shares and make financial decisions accordingly. This study is prepared from the investor's perspective.

Introduction

On the 23rd June 2016, the referendum took place wherein the United Kingdom voted to leave the European Union (EU). After three years of political tug of war, Boris Johnson finally delivered on his promise to deliver Britain's exit from the EU(also referred to as Brexit'), in turn bringing certainty back to British politics and giving businesses and individuals the ability to plan the future with greater clarity. 'Opportunities in the UK have been overlooked for three and a half years because of the political situation... greater clarity really does reduce the level of risk for investors both domestic and international who have found it difficult to assess the prospect of UK equities from afar'(McGagh,2019). In the uncertainty surrounding Brexit, the Pound Sterling fell sharply against its main counterparts including the Euro and US Dollar. This acted as an automatic stabiliser for the UK economy: It made exports relatively cheap and imports expensive. This also made the UK stock market and UK firms in general seem to be

of good value, especially from the perspective of foreign investors ‘the UK stock market is unusually cheap compared to the rest of the world’ (*Somerset Webb,2019*). For this reason, the Financial Times Stock Exchange 100 Index (FTSE 100) has been used as a benchmark for comparing the performance of the company, as Intertek group sell its profits made in foreign currencies for sterling meaning that supernormal profits are stated in the financial statements, mentioned in the appendix. Initially due to Brexit, the company faced little issues in the European market but, now they have relocated their U.K based notified bodies in other EU states

The global product testing and quality assurance market is considered to be worth approximately US\$250 billion. Of this, Intertek estimate that only 20 per cent is currently outsourced. This means that there is an exciting opportunity for Intertek group to capture a share of the US\$200 billion that is currently managed in-house. As per the Conservative party manifesto from the recent election, people hope to see an expansionary fiscal policy in the UK. This increase in government spending will stimulate the economy by boosting aggregate demand. Much of this new Government spending is focused on infrastructure and technology, for example : the building of High Speed 2, one of the most exciting transport projects in Europe. Another conservative pledge is to invest in the parts of England that haven’t seen a much spending as London. These measures will boost demand in the short run by causing a multiplier effect. I believe that this fiscal expansion will lead to an accelerator effect as seen in the U.S (*Duffin.E, 2020*). Spending on infrastructure increases the growth potential of the country as a whole. This view is supported by financial services company *Hargreaves Lansdown* who recently expressed their belief that if executed well, an increase in government spending will put money in people’s pockets and encourage spending which will bode well for UK focused retailers, tech industry, trades, financial services and construction groups(*Hyett,2020*). Also, the government has increased the subsidies for the manufacturing companies who are the main customers of the Intertek group. The increased subsidies for Intertek’s customers led to an increase in GDP as well. For this reason, and as discussed further in this report, Intertek Group has been chosen for this research project. The quality and assurance sector has seen a huge growth over the years and has a never-ending demand because of the rising population, companies and support for quality products.

About the Company :

Intertek is the number 1 company in the U.K and one of the leading companies in the Quality and Assurance sector in the world. The company provides inspection, product testing, assurance and certification of quality to its customers. The company offers its services to different industries like footwear, toys, textiles, building products, industrial, automotive, hardlines, consumer electronics, healthcare, pharmaceuticals, beauty, renewable energy, food and hospitality sector. Intertek market capitalisation at the end of 2019 was £ 9.44 billion. It operates in Europe, Africa, Asia and America. The main competitors of the company are Eurofins Scientific Technology, Bureau Veritas SA (BVI SA) and SGS SA (Société Générale de Surveillance). Eurofins and BVI are traded in European exchange (Euronext) and are based in France operating worldwide whereas, SGS SA is a Swiss multinational company traded on Swiss exchange (SIX). Quality Assurance Industry comprise of these three top companies and that is why they have been chosen as the peer companies of Intertek for comparison purposes.

Across the ATIC (Agricultural Technological Information Centre) market Intertek delivers services like laboratory safety, sustainability analysis, second-party supplier auditing, product assurance, process performance analysis, vendor compliance, quality and performance testing, plant and equipment verification and, third-party certification. Intertek leverages their global network of accredited facilities and world leading technical experts to help their clients meet high quality safety, regulatory and brand standards, develop new products, materials and technologies and ultimately assist them in getting their products to market quicker, in order to continually meet evolving consumer demands. Intertek's unique cyber quality testing and quality assurance program offers continuous, real time vulnerability monitoring and a certification mark to give consumers an unprecedented level of confidence in the security of their device. With Cyber Assured, customers are assured to benefit from the highest level of cyber protection over the lifetime of their products. Intertek's proprietary High-Performance Mark sets the standard for assuring product claims, quality of premium and high-tech textile products in the fast-growing athleisure market. The customers are able to differentiate their athleisure offering, giving consumers the peace-of-mind that their purchase is safe, of high quality. The customers are reassured that the high-performance claims made validated and certified. As part of their Operational Sustainability Services, the company is coming up with new certifications for Recycled Content, Reduced Resources, Carbon Foot-printing and Biodegradability. With independent verification of the sustainability of their products, the

client can effectively manage increased scrutiny from all stakeholders and more stringent regulatory requirements(*by Intertek group,2019*)

The company's Trade division consists of three global business lines with global and regional trade flow based on similar mid- to long-term structural growth drivers. This division focuses on customers from almost every sector :

1. The Caleb Brett business provides cargo inspection, analytical assessment, calibration and related research and technical services to the world's petroleum and biofuels industries.
2. Government & Trade Services('GTS') division provides inspection services to governments and regulatory bodies to support trade activities that help the flow of goods across borders, predominantly in the Middle East, Africa and South America.
3. Lastly the Agri-World business provides analytical and testing services to global agricultural trading companies and growers.

By incorporating the new International Maritime Organization (IMO) 2020 Compliance which allows for a maximum of 0.5% sulphur content in marine fuels globally, Intertek has also attracted a lot of shipping companies as their customers. The Caleb Brett division uses new near-infrared scanners to look outside the usual fuel analysis parameters to identify unexpected issues with oil quality and compatibility. This unique technique allows their shipping customers to mitigate the risks arising from the changes they have made to their fuel to comply with the IMO2020 emissions regulations. Intertek also has a beer-laboratory which delivers remote samples and gives craft brewers access to state-of-the-art lab equipment and total quality assurance expertise. Since one of the most important factors to consider for craft brewing is the taste and smell, the brewers can rapidly identify quality issues, resulting in reduced wastage and optimisation of ingredients for a more efficient brewing process.

The company's services are focused on many sectors and one of them is the quality of construction companies and its product manufacturing industries, comprising of both the private and commercial sector. These industries have observed a rise in value since the Brexit referendum, for example : in June 2016, the average price for a home was £212,887 compared to December 2019 where the average price spiked to £234,742. The capital has seen the biggest hit to property prices since the referendum with 14 quarters of negative growth, this price correction was smaller than the last recessions in the early 1990's and late 2000's. With the fears surrounding Labour government gone, the market is in an upswing, resulting in an increase of customers for the company. The general consensus in the City of London is that

with Boris Johnson now holding office in 10 Downing Street many firms across the sectors have been identified to have the ability to flourish - ‘beneficiaries of a Johnson-led Government would be the domestically exposed mid-caps: these are energy, petroleum, cyber securities (tech industry), construction, utilities, general goods, retail, insurance, food and beverage sector’ (*McGagh, 2018*). Furthermore, housebuilding is at a 30 year high with 241,130 homes built. This trend is set to continue with over 380,000 new homes being built. The government has its own target for 300,000 new homes to be built each year (*Land Registry, 2019*). This will further increase the revenue of the company because Intertek provides assured quality inspection services to the manufacturers of household building materials. The company moves with the market so it is important to analyse the market behaviour.

The second most prominent sector serviced by Intertek group is consumer goods. Consumer spending accounts for about 60% of the British economy and since the Brexit referendum, it has always been growing over at the rate of 1.1% year on year (*ONS, 2019*) and this growth trend can be expected to continue in the UK: ‘I am of the opinion that some of the most attractive opportunities appear to be in the domestic market, strong employment and reasonable wage growth will ease the squeeze on households. Companies serving UK consumers should be among the beneficiaries’ (*Noffke, 2018*). Although the sector has been hit by higher wage costs, overall rising income was ‘a good indicator for consumer spending and the leisure industry’ (*McGagh, 2018*). Intertek is also the number one player in many important markets, for example it is the key player in the most important emerging market in Africa; Nigeria. Nigeria is projected to become the world’s third most populated country by the year 2050 (*UN, 2020*). Meanwhile Intertek clients in the food industry - the leading bakery chain has partnered with the food delivery company named Just Eat to lead the food delivery sector. The two firms have capitalised on food takeaway trends within the industry. Greggs has had a very good year with the opening of over 90 bakeries (*Greggs, 2019*) and furthermore Just Eat is now delivering Greggs after successful trials and will roll out delivery based on local demand. Rising growth in this industry has benefitted Intertek revenue.

Finally, the information sector, a fast-growing sector as the value of information for businesses is increasing. Intertek’s recent acquisitions in 2018-19 has introduced a new line which is in conjunction with their other businesses. This industry is known as the data economy where information is said to be the most valuable asset in the world surpassing oil. Information gathered by the company will improve Intertek management decision making, which could lead to more beneficial outcomes for the Intertek Group such as waste reduction, ability to

reach a new audience or increase revenue. This information could be used to improve strengths, find weaknesses, discover opportunities and mitigate threats.

These new changes happened when Intertek in March 2016, announced a ‘5x5’ transforming strategy for growth, with the aim to move the centre of attraction of the group towards high-growth, high-margin areas in its industry. This approach included two strategic priorities relevant to the operational structure of the business. First, to operate a portfolio that delivers focused growth amongst the business lines, countries and services, including a strategic review of underperforming business units and second, to deliver operational excellence in every operation to drive productivity, which in turn led to various changes in the organization.

Intertek has high earnings retentions (that is 50%) to invest in positive internal rate of return projects with high return on equity. Future potential projects will increase their value, thereby raising the company share prices. This is also beneficial for tax purposes because Intertek is targeting individuals that fall in the high earning bracket. The tax on dividend income for higher rate tax payers is 32.5% and for additional rate payers is 38.1% (*Gov.uk, 2020*) whereas capital gains are only taxed at 20% for additional rate tax payers. This can be offset from the losses if incurred by the company, thereby reducing the money paid to the taxman. With this in mind U.K equities market has been identified as the target market

Mergers and Acquisitions

The Intertek Group completed four (two in 2017) acquisitions by the year 2018, with cash consideration of £387.9m (2017: £27.4m), net of cash acquired of £5.6m (2017: £2.1m). In March 2018, the company acquired Aldo Abela Surveys Limited (“AAS”), a leading provider of quality and quantity cargo inspection services, based in Malta. In April 2018, the firm acquired Proasem S.A.S. (“Proasem”), a leading provider of laboratory testing, inspection, metrology and training services to the Oil & Gas sector, based in Colombia. In June 2018, the firm acquired NTA Monitor Limited (“NTA”), a leading network security and assurance services provider, based in the UK and Malaysia. In August 2018, the firm acquired Alchemy Investment Holdings, Inc. (“Alchemy”), an industry leader in People Assurance solutions for the food industry, based in the USA. In December 2019, the Group acquired Check Safety First Limited, a market leading global health, safety, quality and security risk management business focused on the travel, tourism and hospitality sectors. The Company also invested £116.8m (2018: £113.2m) organically in laboratory expansions, new technologies and equipment and other facilities. This investment represented 3.9% of revenue (2018: 4.0%). These acquisitions

will provide valuable additional service lines and new geographic locations for the company, and will also help drive future profitable revenue, further acceleration of growth momentum of the high-margin and capital-light assurance business. The firm transacts in over 80 currencies across more than 100 countries, revenue and profit are impacted by the currency fluctuations. However, the diversification of the Firm's revenue base provides a partial dilution to this exposure.

Literature Review

Financial Analysis

In order to study the financial performance and position of the company it is important to calculate various ratios like profitability, liquidity and leverage. Also, to measure the company effectiveness, efficiency and investment prospects, certain ratios are described and discussed further in this report like return on assets (ROA), Asset turnover, P/E and Earning per Share(EPS). An analysis of the balance sheet, income statement and cash flow of the Intertek group has been provided, which reflects the position of the company as at 31st Dec, 2019. Intertek's cash flow statements were made according to the International Accounting Standards(IAS) 7 but, in 2019 the company adopted International Financial Reporting Standards (IFRS) 16 due to many reasons. Also, the final accounts of the company were prepared in compliance with U.K GAAP (Generally Accepted Accounting Principles), Article 4 of the IAS company act 2006. Now the company fully follows IFRS as adopted by their European competitors. This make the comparison between the peer companies easy as the valuation of various components like depreciation, amortization, inventory, assets fair value and goodwill will be the same.

IFRS 16requires lessees to recognise a lease liability reflecting future lease payments and a right-of-use asset for lease contracts, subject to limited exceptions for short-term leases and leases of low value assets. The quantitative impact of IFRS 16 on the company had a material impact on its balance sheet as both assets and liabilities have been increased. Also, a material impact on key components within the income statement, resulted in operating lease rental charges to be replaced by depreciation and finance costs. The conversion to IFRS 16 has not affected the underlying commercial performance of Intertek Group, nor its cash flows generated in the year.

Defining Variables :

1. Capital structure and Gearing :

The term 'Gearing' refers to a situation which tells how much of the company's current finances are borne by lenders in comparison to the shareholders. The structure of the company can include share capital, reserve and surplus called shareholders fund or the total debt which comprises of long-term debt. Financial leverage ratio is defined as, total long-term debt divided by total equity, which tells how much proportion of equity and debt is involved in the company. The definition of Gearing has been viewed by different authors in a different way. *Ola (1985)* included fixed interest capital payment and fixed dividend payment to the lenders and ordinary shareholders in the definition of gearing. *Tunji et.al (2015)* included long term debt and payment to preference shareholders in calculating the leverage ratio. In this study Long Term Debt constitutes capital lease and bond payments. The ideal debt to equity ratio is 2.0 which explains that debt should not be employed by more than 66.67% of the total capital. Generally, any ratio below 2.0 is seen as a good business structure. A ratio above 2.0 is viewed as a business risk by the external parties and are of the perception that in the long run the company will face difficulties in clearing off their secured and unsecured loans. The use of debt capital has to be reasonable for a highly geared large size company as the proportion of interest is of significant amount. If there is no other alternative apart from debt for investment purposes and the company falls short in paying the interest in future, then it has to realise its assets (if the debt is secured in any way). Furthermore, high levels of debt could also cause taxpayers money when in financial distress. Therefore, the company has to accumulate enough profits to cover its debt payment otherwise it could lead to bankruptcy.

Intertek has a long-term debt of £ 802.2m which consist of lease payments of international facilities and repayment of senior notes (bonds) issued at 3.2 % in the previous years. Intertek debt to equity ratio is 0.84, which is quite below the ideal ratio. In 2016 this ratio was 1.44 and after that it gradually decreased, showing increased profits. But, in 2018 there was a slight increase in the debt because of the acquisitions made by the company. Intertek debt to equity ratio is also well below the industry average (1.81) and from every other peer company, indicating that the company is less reliant on the debt for its finances as compared its competitors. A study by *Hovakimian et.al (2004)* which uses annual firm level data for the years between 1982 to 2000 found evidence that, profitability has no effect on target leverage because during high operating performance, a company might prefer internal financing rather than external exogenous financing, to maintain the targeted debt ratio and this is why Intertek

has low leverage ratio. The company has a balanced capital structure enabling funding for near growth opportunities.

2. Debt payment

Modigliani & Miller (1963) trade-off theory finds a positive correlation between the profitability and leverage. Profit which arises due to savings of tax (by employing debt), lower probability for bankruptcy, gives more investment to the company future projects. The above view is supported by *Myers's* (2001) study stating, with high profitability the company can service more debt. Intertek capital structure is somewhat related to MM theory, having an optimal level of debt and equity in their balance sheet. In 2017 and 2018 to expand their business Intertek used its retained earnings and partly issued debt to finance their investment. Initially, the net debt increased from £544.1m (Dec,2017) to £778.2m (Dec, 2018), primarily reflecting the acquisition of Alchemy in August 2018. But, in 2019 the company paid its partial debt from its high profits resulting a decrease of £148.8m hence totalling it to 629.4. On the basis of IFRS, Net Debt is £ 802.2m

3. Dividend policy :

The company's dividend pay-out since 2016 has not been constant. It has been increasing at a yearly rate of 14.26%, 39% and from 2018 a change of 6.8% has been marked. In 2019 and 2018 the company announced a dividend of 105.8 and 99.1 respectively, representing a cost of £170.8m and £159.9m which is the 50% of the adjusted profits. The dividend is covered 2 times by the profit. Having a retention rate of 50% from 67% displaying growth, reliability of steady income and confidence in the company's management. A well proportion retention rate is observed as a strategy for investing in future business prospects. According to *Lie* (2000) findings, companies increase dividends because of the excess cash held in comparison to the peer firms in the industry, which is observed in the financial statements mentioned in the appendix section. The dividend yield of Intertek is 1.8% representing a strong position. This ratio explains how much income the shareholders are receiving in relation to the market share price. The average yield of the industry is 1.9 %. Investing in Intertek group is beneficial for long term investors who wants high dividend as the share price is high.

4. Profitability

Myer & Majluf (1986) Pecking-Order model proposes to use profits to finance its operation, the availability of which depends if the leverage is low. Therefore, it predicts a negative relationship between financial leverage and profitability. Also, the Market Timing theory buttressed the earlier study of Myers (1986) by stating that the profitability may be associated with less leverage, thereby relating to internal financing.

To measure profitability, net profit margin ratio (net profit divided by total revenue) has been analysed. Intertek has relatively high margins (10.48%) when compared to other companies and the industry (7.16%). Since 2016, Intertek profits have been increasing. From the year 2018, net income has gone up by 10.09% and profit margins by 3.25%. There was a Net exchange gain/(loss) on hedges of net investments in foreign operations of £ 31.2m thereby pumping up the profits.

5. Liquidity

To assess the company liquidity, current ratio (current assets divided by current liabilities) has been deployed. The ideal ratio is 2:1 which elucidates the company's ability to pay its short term obligations that are due in less than a year. Over the past three years Intertek liquidity (1.07) has decreased. This is due to the repayment of its senior notes, interest and capital leases. But, still there are enough current assets to cover its short term obligations (bank overdraft). The industry average is 1.44 which is neither high nor low and Intertek's liquidity is well within the defined industry average. The company cash and marketable securities are increasing since 2016 even though they have spent hard cash for their acquisition. Also, in 2019 the Group acquired the remaining shares in a non-controlling interest for cash consideration of £5.2m. Prepaid expense has been decreased in response to the increased liquid cash. Bureau Veritas (BVI) and SGS SA have higher liquidity that of 1.67 as compared to Intertek group and this is because these firms are carrying higher debt in their balance sheet.

6. Efficiency

Asset turnover ratio (ATR) is used to test the efficiency of the company. The numerator contains total revenue and the denominator has the average of current year and previous year total assets. The ratio explains how much of the company assets are churned to generate sales. Higher the ratio the better it is for the company. Comparing Intertek Group with the benchmark and other peer companies, it is observed that the company has performed pretty good. Intertek

ATR is 1.10 and the industry average is 0.82. The revenue of the company has been flourishing since 2016 and has seen a 6.6% growth from 2018. These increased sales are due to new demand generated in America, Europe, Asia and in Middle Eastern countries by establishing its base through recent acquisition of 'Check Safety First Limited' who has operations throughout the world.

7. Effectiveness

A company's effectiveness, productivity and performance can be assessed by evaluating the return on assets (ROA), which is calculated by dividing Net Income by Average Total Assets. ROA ratio helps explain how easily and readily the company can convert its invested capital/assets into profits. Higher ratio is what is preferred by the market. However, the company's ROA has decreased over the years because of the increase in purchase of property, plant and equipment, rising debtors and increase in goodwill through acquisition. The increment in assets is much more than the net income because interest payments are rising and operating expenses are going up (due to managing of the large workforce). Yet Intertek group ROA (11.49%) is quite high as compared with other market leaders and the industry (6.28%).

8. Investment prospects :

When reviewing companies for investment purposes most investors look for high earnings, steady income and growth potential. Earning per Share (EPS) and Price to Equity (PE) are the two most important ratios when analyzing the financial statements of the company. EPS ratio translates what the investor can receive for every share they hold. Higher EPS reveals higher earnings for every outstanding share. Also, a steady EPS suggests that the company has potential to pay-out dividends in future. Intertek's EPS is 194p which is very high as compared to their European competitors and the industry (33.32). This is because of higher earnings and achieving market monopoly in the domestic market, the company net income has gone up. In the year 2019, the company also paid a dividend of 108.5p striking a 50% payout policy, exhibiting powerful financial strength.

9. Stock performance:

To solidify ratio analysis and more precisely assisting the strong financial performance of the company and its share price, Abnormal Buy and Hold Return strategy (ABHR) is deployed which is referred to as Event Time Approach. Previous 3-year share prices of Intertek Group and FTSE 100 has been analysed because ABHR measures the long-term stock performance

of the company by comparing it with the market (FTSE 100). According to *Kooli and Suret (2004)*, 'ABHR has an upper hand in measuring the investor experience. In particular, the long-term investor experience is better captured by compounding short-term returns to acquire long-term buy-and-hold returns. Intertek's ABHR over the 36-month analysis is 70.55% which highlights that the firm performed better than the market (FTSE 100). A zero-initial investment in the company would have resulted a profit of 62.55 % in excess of the market index return. Beta is the systematic risk exposing how much the company deviates from the market. Intertek group beta (1.17) is above 1, demonstrating that the company is volatile and moves with the market, displaying higher returns when the FTSE 100 is up.

In terms of investment philosophy, it is appropriate for investors of Intertek Group to take a passive approach focused on long term investment. A top-down analysis strategy is believed to be the most appropriate for Intertek as the UK equities market as a whole is fundamentally undervalued when compared to competitors in Europe. While the UK's top stock market FTSE 100 has a P/E ratio of just 15-16 (*Baxter,2019*), The United States S&P 500 has an average price to earnings ratio of over 22 (*Macrotrends,2020*).As per Fama and French's Five Factor Model, value stocks outperform growth stocks (*Fama and French, 1993*). Upon comparing Intertek group to the market industry, we can see that the company has a price to earnings ratio of 30.16 as compared to 35.08 (the average of the peer companies), which indicates to us that Intertek has been undervalued. Using 'Value investing' method which is followed by investment aficionados such as Warren Buffet and Seth Klarman, it identifies that the chosen company shares are under-priced by the market, indicating that the prices will rise significantly in the future. Essentially this means Intertek shares are trading at a significant discount to their intrinsic value. But, when comparing the company with the benchmark (FTSE100) the P/E ratio is quite high, which states that when the U.K market is bullish Intertek outperforms the market index (*Fisher,1957*).Additionally, across various metrics UK stocks are undeniably competitive 'as a group they (UK stocks) are cheap on valuation measures including price/earnings multiple and dividend yield' (*Noffke, 2018*). Employing a top-down approach tends to create higher turnover than bottom-up approach for investors as there is a greater focus on market cycles rather than individual stocks.

Valuation

In the past, many financial theories and policies have been incorporated by the companies in order to grow their valuation. To have an optimal capital structure that benefits the firm in a right way, a standardised model has never been developed. Capital structure is the way a corporation finances its assets, through a combination of debt, equity, and hybrid securities. Both debt and equity are equally important to the firm. A company fully financed by either of the two cannot last for long because of the frictions in the market. *MM (1963)* highlights the use of debt in order to get tax benefits as interest paid is tax deductible which in turn helps increase the valuation of the enterprise. Clearly 100% debt is not possible due to liquidity, transaction cost, bankruptcy and other such related costs. This is called also called a trade-off model. MM also proposed that the firm dividend policy doesn't affect the firm's market valuation and there is no dependent relationship between them, also elaborated by *Villamil (2000)*.

Valuing a company can be done in many ways, but the three main theoretically accurate primary models are Dividend Discount Model, Earnings and Investment Model and Free Cash Flow Model. When an area is studied as thoroughly and for as long time as valuing a company has been, many competing and complementing theories will inevitably be conceived. Also, in this case, each model offers plausible explanations and justifications of what determines the true value of the business, and what's the best price that can be achieved. The more we read, the more we realise what is missing and that is why this study discusses different methods of valuing a company and the rationale behind any deficiencies of it. In order to maximize the value of the company, both debt and equity is required for the optimal capital structure *Barges (1962)*. The return on capital should be greater than the cost of debt but, for investors to generate excitement, its return on equity should be greater than return on capital

Gordon (1956), **Dividend discount model (DDM)** also called as Gordon Growth Model (GGM) is a method of estimating a company's share price based on the hypothesis that it's share is worth the sum of all of its future dividend payments by discounting of future dividends to their present value. Constant growth rate (g), cost of equity of the company (r) and next year dividend (D) are the determinants required for evaluating the worthiness of the company. According to efficient market hypothesis in a developed economy the share prices reflect all the relevant information of the past, public and of the insides that are affecting the company, directly or indirectly. U.K market has a strong form of efficiency. Therefore, in order to gauge the market and generate profit, one must inspect minutiae intangible factors impacting the

company. The model's main assumptions are that it believes dividend will continue to accrete at a constant level. Also, a company which doesn't pay dividend will have difficulty using this method and will only be able to adopt this approach by speculating the future dividend. Intertek is one of the oldest companies (founded in 1996) and has been paying dividend since 2003. The theoretical value of the firm according to the GGM is £ 2607.80 which is quite below the price quoted in the market (£ 5852). If the value generated by DDM is smaller than the market price, it is assumed that the share price of the company is overvalued. Theoretically it is true but, the model doesn't constitute all the exogenous factors affecting the share price of the company. A detailed explanation of the figures is given in the appendix section.

Earning and investment-based model values the business by speculating its future earnings and investment growth opportunities and bringing them back to its present value. According to (Bell and Edwards 1961, Peasnell 1982, Edey 1962,), 'Popularly the economic value can be expressed as the sum of accounting book value and the present values of all expected future incomes. The variables used in this method are, expected next year (2020) earnings (E_1), growth rate of dividends (g), present value of growth opportunities (PVGO) and the market required rate of return (r). Growth rate of dividends is assumed to be the same for its future value of investments/growth opportunities. The earnings model is derived from Gordon Growth Model. Intertek Group Plc stock price according to this method is £ 2582.72, which is relatively similar to the Dividend Discount Method and well under the actual market price of the share (£ 5852). The present value growth of future investments is zero because the market required rate of return for the company (approx.14%) is greater than the firm investment rate of return (IRR= 13.60%). That is why Intertek dividend pay-out rate increased from 37% of the 2017 adjusted profits to 50% in 2019. *Grullon, Michaely and Swaminathan (2002)* in depth research on dividend pay-outs concludes that, 'firms anticipating negative investment opportunities in the near future are likely to raise dividends. Market Rate of Return for the Intertek Group is derived by taking into account Capital Asset Pricing Model (CAPM), which absorbs the volatility of the company in relation to the market. The risk-free rate of return is 1.51% (Gilt bond), risk/beta of the firm is 1.17 and market risk premium is 10.5%. The future earnings (E_1 ; £ 356.28 m) has been calculated by using the existing dividend growth rate (6.8%) because of the systematic risk in the global market caused by Covid 19. It is likely that the sales and net income would go down next year and will not be able to achieve the past year change in retained earnings that is 10.09%. Further elaboration of this model figures is done in the appendix below.

Free Cash Flow to Firm (FCFF) based valuation Model :

Free cash flow is the cash available to all the security holders (equity, debt and preference shareholders), generated by the core business operations. It is the liquidity that is available after covering up capital expenditure and net working capital. This strategy of computing the worthiness of the company involves discounting future cash flows to its present value by using weighted average cost of capital (WACC). The definition of WACC is mentioned in the appendix section. An essential assumption for this approach is that the cash flows should grow at a constant pace. From Free Cash Flow method, Intertek is valued at £ 2789.60. The company's interest expense on the debt for the year 2019 is £ 40.7m (2018; £ 29.3m). The rate at which the fixed interest payments are given is called the cost of debt which is tax deductible. This tax shield helps firms to maximize their value. Return on Equity on the other hand is the Rate of Return earned by the top shareholders/investors of the company. The return on equity of Intertek is 34.90% which is high because of the risk borne by the investors. The drawbacks in this process are the hidden figures which the company does not disclose, it provides a rough estimate of the values but, does not reveal the details. Due to this, the value generated by FCFF Model is not equal to the Dividend Discount Model but, the figures prove that the market price of the share is overvalued, similar to the projections of DDM and Earning Based Model. This might not be true because of the hindrances faced by the theoretical approaches like speculation of future growth rate, systematic risk, firm risk and limitations of accounting policies, etc.

Conclusion

Through this research, it can be seen that there are conflicting results in terms of financial position and theoretical valuation of the Intertek Group. According to the distinctive primary valuation models used in this study, it is reckoned that the company as on 31st December 2019 is overvalued because the share price on that date was £5852. This increase in value may be due to company buying back its own shares for the total of value £23.1m. Other intangible factors could also play a role but, the price equity ratio explains a different side of the story. Intertek P/E is below the industry average which illustrates that the company is a value-based company and the price would surge further in the long run. Also, the ratios and the final accounts of the Group show an increase in net income, revenue and the assets. In comparison to other peer companies, Intertek is most efficient in managing its assets and very effective in generating higher profits on a yearly basis. A steady growth in dividend and share prices reflect the positivity in company. The capital structure of the business is assumed to be based on *Modigliani and Miller (1963)* hypothesis. A debt equity ratio of 0.84 in the fiscal year of 2019

reveals that the company prefers retained earnings over debt, therefore minimising interest cost and the use of ideal amount of borrowing is to get the corporate tax benefits.

The company is observed to follow U.K market (FTSE 100). An ABHR of 62.5% portrays that the return on the company shares is greater than the return on Fointsie (FTSE100). This also explains that when the U.K market is in upswing, the company stocks are also up and vice versa. Intertek company has shown a strong financial performance in the past. The company has introduced new business lines, shifted its focus to different sectors to diversify its business, targeting high growth-margin and expensive industries. Through acquisitions, the company has become one of the world's best product testing and quality assurance company. A 25% increment in the market share has been recorded within a year. Last year, the company's return on shares was 21.5% whereas the return on market was 12% and the required rate of return on the company was 14% (considering the systematic risk in the market). U.K govt stability and fiscal expenditure in the previous year has escalated the GDP and generated a great demand for the company. The firm is operating in an oligopoly market where there are only few sellers for this kind of service, thereby feeding the entire market and cashing out a huge profit from it. The company is benefitted from having a niche segment because of the entry barriers imposed by the market such as licencing, professional expertise, cost, advanced technology and regulatory requirements. The share price may not increase much in the year 2020 because of the worldwide unprecedented risk faced by the market caused by the novel coronavirus. However, if a buy and hold strategy is adopted by investors, the shareholders will be benefitted by the company in the long run, meanwhile they can enjoy a steady income of dividend. In my opinion, Intertek Group is a company with minimal risk, huge upside growth potential and is definitely worthy of investment.

Appendix

The following data has been taken from Yahoo Finance and Intertek Group Annual reports.

Financial Statements

Table 2.1 - Consolidated Intertek Group Plc Balance Sheet

Intertek Balance Sheet				
Period Ending:	31-12-2019	31-12-2018	31-12-2017	31-12-2016
	£ mil	£ mil	£ mil	£ mil
Total Current Assets	960.1	929.3	814.3	869.5
Cash and Short Term Investments	227.4	206.9	137	175.6
Total Receivables, Net	679.1	667.3	628.6	555.8
Accounts Receivables - Trade, Net	584.7	584.9	560	472.8
Total Inventory	19.2	18.3	18.3	19.1
Prepaid Expenses	34.4	36.8	30.4	118.9
Other Current Assets, Total	-	-	-	0.1
Total Assets	2818.4	2633.6	2052.4	2146.3
Property/Plant/Equipment, Total - Net	644.2	441.2	420.6	443.3
Property/Plant/Equipment, Total - Gross	1764.3	1281.9	1182	1177.9
Accumulated Depreciation, Total	-1120.1	-840.7	-761.4	-734.6
Goodwill, Net	859.8	874.9	579.6	586.1
Intangibles, Net	302.4	329.5	178.2	198.8
Long Term Investments	0.3	0.3	0.3	0.3
Other Long Term Assets, Total	51.9	58.4	59.4	48.3
Total Current Liabilities	900	742.7	608.3	600
Accounts Payable	163.8	154	126.7	107.3
Accrued Expenses	240.6	240	227.1	274
Notes Payable/Short Term Debt	14.4	3.7	1.1	16.8
Current Port. of LT Debt/Capital Leases	286.2	134.6	76	86.6
Other Current liabilities, Total	195	210.4	177.4	115.3
Total Liabilities	1862.5	1759.6	1342.7	1578.6
Total Long Term Debt	802.2	846.8	604	815.9
Long Term Debt	617.9	846.8	604	815.9
Capital Lease Obligations	184.3	-	-	-
Deferred Income Tax	68.2	80.8	47.4	48.7
Minority Interest	29.4	34.3	34.5	34.7
Other Liabilities, Total	62.7	55	48.5	79.3
Total Equity	955.9	874	709.7	567.7
Common Stock, Total	1.6	1.6	1.6	1.6
Additional Paid-In Capital	257.8	257.8	257.8	257.8
Retained Earnings (Accumulated Deficit)	696.5	614.6	450.3	308.3
Total Liabilities & Shareholders' Equity	2818.4	2633.6	2052.4	2146.3
Total Common Shares Outstanding	161.39	161.39	161.39	161.39

Table 2.2

Consolidated Intertek group Income Statement

Income statement (IFRS 16)				
Period Ending:	31-12-2019	31-12-2018	31-12-2017	31-12-2016
	£ mil	£ mil	£ mil	£ mil
Total Revenue	2987	2801.2	2769.1	2567
Revenue	2987	2801.2	2769.1	2567
Total Operating Expenses	2502.5	2371.4	2346.4	2197.5
Depreciation / Amortization	29.1	24.6	16	14
Interest Expense (Income) - Net Operating	0.8	-	-	-
Unusual Expense (Income)	9.8	27.4	29	26.2
Other Operating Expenses, Total	2462.8	2319.4	2301.4	2157.3
Operating Income	484.5	429.8	422.7	369.5
Interest Income (Expense), Net Non-Operating	-39.4	-25.3	-29.4	-22.4
Net Income Before Taxes	445.1	404.5	393.3	347.1
Provision for Income Taxes	111.5	99.3	99.4	75.5
Net Income After Taxes	333.6	305.2	293.9	271.6
Minority Interest	-20.5	-20.8	-19	-16.6
Net Income Before Extraordinary Items	313.1	284.4	274.9	255
Total Extraordinary Items	-	-	12.5	-
Net Income	313.1	284.4	287.4	255

Table 2.3- Consolidated Intertek Group Cash Flow statement

Consolidated Cash flow Statement				
Particulars	2019	2018	2017	2016
	£m	£m	£m	£m
Cash flows from operating activities				
Profit for the year	333.6	305.2	306.4	271.6
Adjustments for:				
Depreciation charge	156.2	76.2	81.2	76.4
Amortisation of software	15.3	12.5	12.2	13.1
Amortisation of acquisition intangibles	29.1	24.6	16	14
Amortisation of acquisition intangibles	–	–	18.2	–
Equity-settled transactions	21.9	20.9	17.5	16.6
Net financing costs	40.7	31.7	29.4	22.4
Income tax expense	111.5	99.3	86.9	75.5
Profit on disposal of subsidiary/associate	-1.8	-1.1	–	2
(Profit)/loss on disposal of property, plant, equipment and software	-0.9	0.4	-0.8	-0.1
Operating cash flows before changes in working capital and operating provisions	705.6	569.7	567	491.5
Change in inventories	-1.5	1	-0.5	–
Change in trade and other receivables	-25.6	-16	-22.7	28.8
Change in trade and other payables	40.7	35.2	46	21.9
Change in provisions	-1.9	-7	-7.8	4
Special contributions into pension schemes	-2	-2	-2.8	-2.8
Cash generated from operations	715.3	580.9	579.2	543.4
Interest and other finance expense paid	-40.7	-29.3	-28.3	-29.7
Income taxes paid	-111.8	-93.1	-100.8	-94.1
Net cash flows generated from operating activities	562.8	458.5	450.1	419.6
Cash flows from investing activities				
Proceeds from sale of property, plant, equipment and software	2.5	3.5	3.2	3
Interest received	1.2	1.8	1.2	1
Acquisition of subsidiaries, net of cash acquired	-16.9	-387.9	-27.4	-34.8
Consideration (paid)/received in respect of prior year acquisitions	-0.6	0.1	-7.8	-2
Sale of associate	2.1	–	–	-1.4
Acquisition of property, plant, equipment and software	-116.8	-113.2	-112.9	-105.5
Net cash flows used in investing activities	-128.5	-495.7	-143.7	-139.7
Cash flows from financing activities				
Purchase of own shares	-23.1	-16.7	-15.6	-6.4
Tax paid on share awards vested	-11.6	-9.9	-6.8	-5.2
Drawdown of borrowings	110	341.4	–	0.2
Repayment of borrowings	-221.3	-75.9	-151.3	-170.5
Repayment of lease liabilities	-69.7	–	–	–
Purchase of non-controlling interest	-5.2	–	–	–
Dividends paid to non-controlling interest	-19.1	-18.2	-18.7	-16.3
Equity dividends paid	-163.2	-128.3	-107	-88
Net cash flows (used in)/generated from financing activities	-403.2	92.4	-299.4	-286.2
Net increase in cash and cash equivalents	31.1	55.2	7	-6.3
Cash and cash equivalents at 1 January	203.2	135.9	158.8	116
Exchange adjustments	-21.3	12.1	-29.9	49.1
Cash and cash equivalents at 31 December	213	203.2	135.9	158.8

Peer Companies Financial Statements

Bureau Veritas

Table 2.4 – Consolidated Balance Sheet of Bureau Veritas

Bureau Veritas Balance Sheet				
Period Ending:	31-12-2019	31-12-2018	31-12-2017	31-12-2016
Particulars	£ mil	£ mil	£ mil	£ mil
Total Current Assets	3298.6	2728.9	1987.8	2693.8
Cash and Short Term Investments	1501.2	1059.4	384.6	1145.1
Cash	1046.5	438.8	357.2	425.4
Cash & Equivalents	431.3	607.5	7.1	668.7
Short Term Investments	23.4	13.1	20.3	51
Total Receivables, Net	1737	1645.8	1579.5	1524.4
Accounts Receivables - Trade, Net	1254.4	1201.7	1130.5	1324.4
Total Inventory	56	19.9	19.9	20.6
Prepaid Expenses	-	-	-	-
Other Current Assets, Total	4.4	3.8	3.8	3.7
Total Assets	7049.1	6096.3	5350.2	6095.2
Property/Plant/Equipment, Total - Net	813.9	471.1	486.3	518.6
Property/Plant/Equipment, Total - Gross	1862.8	1364.5	1325.6	1355.7
Accumulated Depreciation, Total	-1048.9	-893.4	-839.3	-837.1
Goodwill, Net	2075.1	2011.6	1965.1	1977.6
Intangibles, Net	611.1	634.6	640.2	686.8
Long Term Investments	118.3	114.8	124.3	75.5
Other Long Term Assets, Total	132.1	135.3	146.5	142.9
Total Current Liabilities	1978.9	1883.2	1526.2	1811.7
Accounts Payable	244.1	390	372.8	347.9
Accrued Expenses	581.2	572	541.5	501.2
Notes Payable/Short Term Debt	12.1	11.7	9.8	6
Current Port. of LT Debt/Capital Leases	449.5	487.3	199.2	583.5
Other Current liabilities, Total	692	422.2	402.9	373.1
Total Liabilities	5785.3	5137	4379.8	4897.8
Total Long Term Debt	3244.5	2655.7	2240	2567.7
Long Term Debt	2918.5	2655.7	2240	2567.7
Capital Lease Obligations	326	-	-	-
Deferred Income Tax	122.9	127.4	143.3	164.8
Minority Interest"	58.3	48.3	42.7	45.6
Other Liabilities, Total"	380.7	422.4	427.6	308
Total Equity	1263.8	959.3	970.4	1197.4
Common Stock, Total	54.2	53	53	53
Retained Earnings (Accumulated Deficit)	1209.6	906.3	917.4	1144.4
Total Liabilities & Shareholders' Equity	7049.1	6096.3	5350.2	6095.2
Total Common Shares Outstanding	447.7	435.91	436.02	436.73

Table 2.5 – Consolidate Bureau Veritas Income Statement

Bureau Veritas Income Sheet				
Period Ending:	31-12-2019	31-12-2018	31-12-2017	31-12-2016
Particulars	£ mil	£ mil	£ mil	£ mil
Total Revenue	5099.7	4795.5	4689.4	4549.2
Revenue	5099.7	4795.5	4689.4	4549.2
Other Revenue, Total	-	-	-	-
Cost of Revenue, Total	1374.4	1273.4	1243.8	1196.1
Gross Profit	3725.3	3522.1	3445.6	3353.1
Total Operating Expenses	4378.4	4158.3	4083.1	3939.5
Selling/General/Admin. Expenses, Total	2706.5	2697.9	2645.7	2538.9
Depreciation / Amortization	305.3	200.3	203.7	202.4
Interest Expense (Income) - Net Operating	-	-	-	-
Unusual Expense (Income)	1.4	3.6	2.2	1.7
Other Operating Expenses, Total	-9.2	-16.9	-12.3	0.4
Operating Income	721.3	637.2	606.3	609.7
Interest Income (Expense), Net Non-Operating	-109.6	-87.8	-98.3	-80.4
Other, Net	-8.4	-5	-4.8	-5.3
Net Income Before Taxes	603.3	544.4	503.2	524
Provision for Income Taxes	210.7	189.3	164.8	188.9
Net Income After Taxes	392.6	355.1	338.4	335.1
Minority Interest	-24.7	-22.5	-21.8	-15.7
Net Income Before Extraordinary Items	367.9	332.6	316.6	319.4
Total Extraordinary Items	-	-	-8.6	-
Net Income	367.9	332.6	308	319.4

Table 2.6 – Consolidate Bureau Veritas Cash Flow Statement

Bureau Veritas Cash Flow Statement				
Period Ending:	31-12-2019	31-12-2018	31-12-2017	31-12-2016
Particulars	£ mil	£ mil	£ mil	£ mil
Net Income/Starting Line	603.3	544.4	503.2	524
Cash From Operating Activities	820.4	685.5	581.4	594.4
Depreciation/Depletion	305.2	200.3	203.7	202.4
Non-Cash Items	121.5	113.2	103.5	119
Cash Taxes Paid	192.4	176.5	169.7	213.8
Cash Interest Paid	79.8	83	98.2	86
Changes in Working Capital	-209.6	-172.4	-229	-251
Cash From Investing Activities	-194.2	-274.9	-312.1	-324.9
Capital Expenditures	-127.9	-130.9	-142.3	-156.6
Other Investing Cash Flow Items, Total	-66.3	-144	-169.8	-168.3
Cash From Financing Activities	-193.6	278	-975.1	310.4
Financing Cash Flow Items	-79.8	-83	-98.5	-86
Total Cash Dividends Paid	-97.3	-277.7	-295.4	-255.1
Issuance (Retirement) of Stock, Net	17.6	-28.3	-33.4	-41.8
Issuance (Retirement) of Debt, Net	-34.1	667	-547.8	693.3
Foreign Exchange Effects	-1.5	-8.5	-27.7	-2.6
Net Change in Cash	431.1	680.1	-733.5	577.3
Beginning Cash Balance	1046.7	366.2	1097.8	510.8
Ending Cash Balance	1477.8	1046.3	364.3	1097.8
Free Cash Flow	692.5	554.6	439.1	437.8

Eurofins Technology

Table 2.7 – Consolidated Balance Sheet of Eurofins Tech.

Eurofins Technology Balance Sheet				
Period Ending:	31-12-2019	31-12-2018	31-12-2017	31-12-2016
Particulars	£ mil	£ mil	£ mil	£ mil
Total Current Assets	1604.1	1705.4	1826.58	1579.41
Cash and Short Term Investments	297	506.25	820.36	826.1
Cash	293.4	24.6	153.29	506.02
Cash & Equivalents	3.6	481.65	667.06	320.08
Short Term Investments	-	-	90.48	85.55
Total Receivables, Net	1185	1043.32	826.8	605.78
Accounts Receivables - Trade, Net	810.1	704.53	576.55	431.63
Total Inventory	79.3	66.37	51.8	39.55
Prepaid Expenses	42.6	42.67	37.14	22.43
Other Current Assets, Total	0.2	46.8	90.48	85.55
Total Assets	7823.1	7205.45	5832.31	4129.41
Property/Plant/Equipment, Total - Net	1593.5	1018	686	506.82
Property/Plant/Equipment, Total - Gross	3171.5	2151.32	1661.55	1242.93
Accumulated Depreciation, Total	-1578	-1133.31	-975.55	-736.11
Goodwill, Net	3608.8	3418.33	2505.34	1584.64
Intangibles, Net	918.2	946.92	735.12	383.21
Long Term Investments	54.5	72.63	49.58	37.53
Other Long Term Assets, Total	44	44.17	29.69	37.8
Total Current Liabilities	1581.6	1450.66	1451.89	710.01
Accounts Payable	231.5	231.94	206.14	230.12
Payable/Accrued	409.8	-	301.86	-
Accrued Expenses	623.7	560.8	435.63	278.76
Notes Payable/Short Term Debt	319.5	372.24	250.33	37.8
Current Port. of LT Debt/Capital Leases	135.3	18.83	303.9	5.72
Other Current liabilities, Total	271.6	266.84	255.89	157.61
Total Liabilities	4984.5	4536.26	3389.14	2501.94
Total Long Term Debt	3086.9	2766.17	1662.1	1340.36
Long Term Debt	2646.8	2727.22	1651.54	1331.89
Capital Lease Obligations	440.1	38.95	10.56	8.47
Deferred Income Tax	124.5	138.56	126.35	83.91
Minority Interest"	59.5	52.99	38.8	129.24
Other Liabilities, Total"	132	127.88	109.99	238.42
Total Equity	2838.6	2669.19	2443.17	1627.48
Common Stock, Total	1.8	1.77	1.76	1.69
Additional Paid-In Capital	-	-	-	-
Retained Earnings (Accumulated Deficit)	1697.1	1614.03	1457.74	902.21
Treasury Stock - Common	-0.2	-0.18	-	-
Other Equity, Total	1139.9	1053.57	983.67	723.58
Total Liabilities & Shareholders' Equity	7823.1	7205.45	5832.31	4129.41
Total Common Shares Outstanding	17.9	17.75	17.64	16.93

Table 2.8 – Eurofins Tech. Consolidated Income Statement

Eurofins Technology Income Sheet				
Period Ending:	31-12-2019	31-12-2018	31-12-2017	31-12-2016
Particulars	£ mil	£ mil	£ mil	£ mil
Total Revenue	4562.8	3781.15	2971.42	2536.61
Revenue	4562.8	3781.15	2971.42	2536.61
Other Revenue, Total	-	-	-	-
Cost of Revenue, Total	3632.1	3061.3	2414.69	2056.98
Gross Profit	930.7	719.85	556.72	479.62
Total Operating Expenses	3989.3	3260.39	2571.86	2178.99
Depreciation / Amortization	357.2	199.09	157.16	122.01
Operating Income	573.5	520.76	399.56	357.62
Interest Income (Expense), Net Non-Operating	-94.4	-59.72	-42.74	-68.7
Gain (Loss) on Sale of Assets	-	-	-	-
Other, Net	-1.9	0	-	19.91
Net Income Before Taxes	477.2	461.04	356.82	308.83
Provision for Income Taxes	116.9	104.98	54.48	81.36
Net Income After Taxes	360.3	356.06	302.34	227.47
Minority Interest	-	0.36	-2.07	-3.51
Net Income Before Extraordinary Items	360.3	356.42	300.27	223.96
Total Extraordinary Items	-165	-132.48	-83.5	-49.96
Net Income	195.3	223.94	216.77	174

Table 2.9 – Eurofins Tech Consolidated Cash Flow Statement

Eurofins Technology Cash Flow Sheet				
Period Ending:	31-12-2019	31-12-2018	31-12-2017	31-12-2016
Particulars	£ mil	£ mil	£ mil	£ mil
Net Income/Starting Line	271.5	289.84	250.1	242.61
Cash From Operating Activities	677.8	543.89	405.12	371.84
Depreciation/Depletion	395.9	223.75	175.72	141.7
Amortization	100	77.81	40.77	-
Non-Cash Items	69.6	41.12	28.64	67.41
Cash Taxes Paid	95	67.82	79.38	73.24
Cash Interest Paid	93.3	59.98	45.61	55.22
Changes in Working Capital	-159.2	-88.62	-90.12	-79.88
Cash From Investing Activities	-540.2	-1557.45	-1747.09	-391.8
Capital Expenditures	-425.1	-364.45	-245.74	-199.68
Other Investing Cash Flow Items, Total	-115.1	-1193	-1501.35	-192.13
Cash From Financing Activities	-342.2	691.06	1343.29	49.9
Financing Cash Flow Items	-461.6	-108.71	317.31	-90.85
Total Cash Dividends Paid	-51.4	-42.65	-34.91	-22.45
Issuance (Retirement) of Stock, Net	23.4	20.04	320.05	501.12
Issuance (Retirement) of Debt, Net	147.4	822.37	740.85	-337.91
Foreign Exchange Effects	4.1	1.47	-10.97	4.67
Net Change in Cash	-200.5	-321.02	-9.64	34.62
Beginning Cash Balance	497.5	827.27	829.99	791.052
Ending Cash Balance	297	506.25	820.35	829.99
Free Cash Flow	252.7	179.44	159.38	172.168

Société Générale de Surveillance (SGS SA)

Table 2.10 – SGS SA Consolidated Balance Sheet

Société Générale de Surveillance Balance Sheet				
Period Ending:	31-12-2019	31-12-2018	31-12-2017	31-12-2016
Particulars	£ mil	£ mil	£ mil	£ mil
Total Current Assets	2964	3301	3140	2611
Cash and Short Term Investments	1475	1751	1393	984
Cash	-	40	40	20
Cash & Equivalents	1475	1702	1343	955
Short Term Investments	-	9	10	9
Total Receivables, Net	1249	1261	1579	1521
Accounts Receivables - Trade, Net	953	1037	1326	1252
Total Inventory	240	272	152	103
Prepaid Expenses	-	-	-	-
Other Current Assets, Total	-	17	16	3
Total Assets	6327	6068	5943	5349
Property/Plant/Equipment, Total - Net	1537	969	1002	972
Property/Plant/Equipment, Total - Gross	-	3337	3287	3023
Accumulated Depreciation, Total	-	-2368	-2285	-2051
Goodwill, Net	-	1224	1238	1195
Intangibles, Net	1468	202	222	246
Long Term Investments	-	36	36	38
Note Receivable - Long Term	-	19	9	14
Other Long Term Assets, Total	358	317	296	273
Total Current Liabilities	1778	1975	1581	1488
Accounts Payable	638	362	344	300
Accrued Expenses	-	618	624	562
Notes Payable/Short Term Debt	-	-	-	1
Current Port. of LT Debt/Capital Leases	192	378	1	-
Other Current liabilities, Total	948	617	612	625
Total Liabilities	4813	4400	4024	3576
Total Long Term Debt	2689	2112	2090	1719
Long Term Debt	2689	2110	2089	1718
Capital Lease Obligations	-	2	1	1
Deferred Income Tax	-	30	45	42
Minority Interest	81	75	86	80
Other Liabilities, Total	265	208	222	247
Total Equity	1514	1668	1919	1773
Common Stock, Total	-	8	8	8
Additional Paid-In Capital	-	-	-	-
Retained Earnings (Accumulated Deficit)	-	3154	3197	3427
Treasury Stock - Common	-	-191	-125	-478
Other Equity, Total	1514	-1303	-1161	-1184
Total Liabilities & Shareholders' Equity	6327	6068	5943	5349
Total Common Shares Outstanding	7.48	7.55	7.55	7.54

Table 2.11 – SGS SA Consolidated Income Statement

Société Générale de Surveillance Income Sheet				
Period Ending:	31-12-2019	31-12-2018	31-12-2017	31-12-2016
Particulars	£ mil	£ mil	£ mil	£ mil
Total Revenue	6600	6706	6349	5985
Revenue	6600	6706	6349	5985
Other Revenue, Total	-	-	-	-
Cost of Revenue, Total	386	387	394	368
Gross Profit	6214	6319	5955	5617
Total Operating Expenses	5518	5760	5455	5169
Selling/General/Admin. Expenses, Total	4688	4745	4437	4158
Research & Development	-	-	-	-
Depreciation / Amortization	487	316	309	305
Unusual Expense (Income)	-205	1	29	31
Other Operating Expenses, Total	162	311	286	307
Operating Income	1082	946	894	816
Interest Income (Expense), Net Non-Operating	-62	-36	-40	-42
Other, Net	-3	-2	-3	-3
Net Income Before Taxes	1017	908	851	771
Provision for Income Taxes	315	218	187	185
Net Income After Taxes	702	690	664	586
Minority Interest	-42	-47	-43	-43
Net Income Before Extraordinary Items	660	643	621	543
Total Extraordinary Items	-	-	-	-
Net Income	660	643	621	543

Table 2.12 –SGS SA Consolidated Cash Flow Statement

Société Générale de Surveillance Cash Flow Statement				
Period Ending:	31-12-2019	31-12-2018	31-12-2017	31-12-2016
Particulars	£ mil	£ mil	£ mil	£ mil
Net Income/Starting Line	702	690	664	586
Cash From Operating Activities	1149	1074	987	1014
Depreciation/Depletion	430	258	257	253
Amortization	57	58	52	52
Deferred Taxes	-	-	-	-
Non-Cash Items	269	238	256	255
Cash Receipts	-	-	-	-
Cash Payments	-	-	-	-
Cash Taxes Paid	306	265	241	207
Cash Interest Paid	261	60	56	58
Changes in Working Capital	-309	-170	-242	-132
Cash From Investing Activities	-100	-314	-311	-201
Capital Expenditures	-290	-304	-298	-289
Other Investing Cash Flow Items, Total	190	-10	-13	88
Cash From Financing Activities	-1303	-370	-239	-1273
Financing Cash Flow Items	-316	-105	-95	-101
Total Cash Dividends Paid	-589	-573	-528	-517
Issuance (Retirement) of Stock, Net	-23	-93	13	-161
Issuance (Retirement) of Debt, Net	-375	401	371	-494
Foreign Exchange Effects	-23	-30	-29	-55
Net Change in Cash	-277	360	408	-515
Beginning Cash Balance	1742	1382	975	1490
Ending Cash Balance	1465	1742	1383	975
Free Cash Flow	859	770	689	725

Ratio Analysis

Calculations

1. Current Ratio = Total Current Assets / Total Current Liabilities
2. Net Profit Margin = Net income after Interest and Taxes / Total Revenue
3. Debt to Equity Ratio = Total Long-Term Debt / Total Equity
 Total Long-term Debt = Long Term Debt/ Bond Holders + Capital lease obligations
4. Return on Assets Ratio = Net income after Interest and Taxes / Average Total Assets.
 Average Total Assets =
$$\frac{\text{Current Year Total Assets} + \text{Previous Year Total Assets}}{2}$$
5. Asset Turnover Ratio = Total Revenue / Avg. Total Assets
6. Basic Earnings per Share =
$$\frac{\{\text{Earnings Before Interest and Tax (EBIT)} - \text{Interest}\} \{1 - \text{Tax Rate}\}}{\text{Number of Shares Outstanding}}$$
7. Price to Equity Ratio (P/E) = Market price of the share / EPS

Some Essential Notes:

- For EPS computation market price of the share is the closing price recorded on 31st Dec, 2019, derived from Yahoo Finance
- In EPS, Earnings = {EBIT-Interest} {1-T_C}
- Tax rate used for Intertek in the year 2019 is 25.1%, reported by the Intertek Group Plc in their annual statement. Also, this rate is consistently the same from the past 4 years

Table 2.13

Metric	Measures	Intertek Ratios			
		31-12-2019	31-12-2018	31-12-2017	31-12-2016
Current Ratio	Liquidity	1.07	1.25	1.34	1.45
Net profit margin	Profitability	0.10	0.10	0.10	0.10
Debt to Equity Ratio	Gearing	0.84	0.97	0.85	1.44
Return on Assets ratio	Effectiveness	0.11	0.12	0.14	0.13
Asset Turnover Ratio	Efficiency	1.10	1.20	1.32	1.30
Basic EPS	Investment Prospects	194.00	207.00	194.10	158.50
P/E ratio	Earnings Multiple/ Valuation	30.16	23.19	26.74	21.96

Table 2.14

Metric	Measures	Bureau Veritas (BVI) Ratios			
		31-12-2019	31-12-2018	31-12-2017	31-12-2016
Current Ratio	Liquidity	1.67	1.45	1.30	1.49
Net profit margin	Profitability	0.07	0.07	0.07	0.07
Debt to Equity Ratio	Gearing	2.57	2.77	2.31	2.14
Return on Assets ratio	Effectiveness	0.06	0.06	0.05	0.06
Asset Turnover Ratio	Efficiency	0.78	0.84	0.82	0.81
Basic EPS	Investment Prospects	0.82	0.76	0.71	0.73
P/E ratio	Earnings Multiple/ Valuation	28.31			

Table 2.15

Metric	Measures	Eurofins Technology Ratios			
		31-12-2019	31-12-2018	31-12-2017	31-12-2016
Current Ratio	Liquidity	1.01	1.18	1.26	2.22
Net profit margin	Profitability	0.04	0.06	0.07	0.07
Debt to Equity Ratio	Gearing	1.09	1.04	0.68	0.82
Return on Assets ratio	Effectiveness	0.03	0.03	0.04	0.04
Asset Turnover Ratio	Efficiency	0.61	0.58	0.60	0.65
Basic EPS	Investment Prospects	10.91	12.62	12.29	10.28
P/E ratio	Earnings Multiple/ Valuation	45.30			

Table 2.16

Metric	Measures	Société Générale de Surveillance(SGS SA) Ratios			
		31-12-2019	31-12-2018	31-12-2017	31-12-2016
Current Ratio	Liquidity	1.67	1.67	1.99	1.75
Net profit margin	Profitability	0.10	0.10	0.10	0.09
Debt to Equity Ratio	Gearing	1.78	1.27	1.09	0.97
Return on Assets ratio	Effectiveness	0.11	0.11	0.11	0.10
Asset Turnover Ratio	Efficiency	1.06	1.12	1.12	1.06
Basic EPS	Investment Prospects	88.24	0.11	0.10	0.10
P/E ratio	Earnings Multiple/ Valuation	31.63			

Table 2.17

Metric	Measures	Benchmark/ Peer Companies Average			
		31-12-2019	31-12-2018	31-12-2017	31-12-2016
Current Ratio	Liquidity	1.45	1.43	1.52	1.82
Net profit margin	Profitability	0.07	0.07	0.08	0.08
Debt to Equity Ratio	Gearing	1.81	1.69	1.36	1.31
Return on Assets ratio	Effectiveness	0.06	0.07	0.07	0.07
Asset Turnover Ratio	Efficiency	0.82	0.84	0.85	0.84
Basic EPS	Investment Prospects	33.32			
P/E ratio	Earnings Multiple/ Valuation	35.08			

Table 2.18-

Dividend Yield as at 31st December 2019

- Average Dividend yield is the industry average which inculcates Bureau Veritas, Eurofins Tech and SGS SA

	Dividend Yield of the Companies				
	Intertek	Bureau Veritas (BVI)	Eurofins Technology	Société Générale de Surveillance(SGS SA)	Average
Divided per share	105.8	0.56	2.88	78	
Share price as at 31st Dec, 2019	5852	23.53	498	2651	
Divided Yield	1.81%	2.38%	0.58%	2.94%	1.97%

Table 2.19

The table below displays Abnormal Buy and Hold Return data of 36 months and Beta of the company.

Calculations :

- Daily return = $\frac{\text{current day stock} - \text{previous day stock}}{\text{Previous day stock}}$

$ABHR_i = \prod_{t=1}^N (1 + r_{i,t}) - \prod_{t=1}^N (1 + r_{\text{benchmark},t})$, Intertek daily product return is measured against the benchmark (that is the FTSE 100)

- Company risk/ β (Beta) = $\frac{\text{Covariance}}{\text{Variance}}$

Covariance is the measure of Intertek stock return relative to the market return (FTSE 100)

Variance is the volatility of FTSE 100

07-10-2019	5330	7197.88	0.004523181	0.005939587	1.004523	1.00594		
08-10-2019	5336	7143.15	0.001125704	-0.007603628	1.001126	0.992396		
09-10-2019	5342	7166.5	0.001124438	0.003268866	1.001124	1.003269		
10-10-2019	5280	7186.36	-0.01160614	0.002771227	0.988394	1.002771		
11-10-2019	5250	7247.08	-0.005681818	0.00844934	0.994318	1.008449		
14-10-2019	5188	7213.45	-0.011809524	-0.00464049	0.98819	0.99536		
15-10-2019	5188	7211.64	0	-0.00025092	1	0.999749		
16-10-2019	5104	7167.95	-0.01619121	-0.006058261	0.983809	0.993942		
17-10-2019	5030	7182.32	-0.014498433	0.002004757	0.985502	1.002005		
18-10-2019	4934	7150.57	-0.019085487	-0.004420577	0.980915	0.995579		
21-10-2019	4950	7163.64	0.003242805	0.001827826	1.003243	1.001828		
22-10-2019	4995	7212.49	0.009090909	0.006819159	1.009091	1.006819		
23-10-2019	4971	7260.74	-0.004804805	0.006689784	0.995195	1.00669		
24-10-2019	5066	7328.25	0.019110843	0.00929795	1.019111	1.009298		
25-10-2019	5142	7324.47	0.015001974	-0.000515812	1.015002	0.999484		
28-10-2019	5262	7331.28	0.023337223	0.00092976	1.023337	1.00093		
29-10-2019	5196	7306.26	-0.012542759	-0.003412774	0.987457	0.996587		
30-10-2019	5328	7330.78	0.025404157	0.003356026	1.025404	1.003356		
31-10-2019	5352	7248.38	0.004504505	-0.011240277	1.004505	0.98876		
01-11-2019	5358	7302.42	0.001121076	0.007455459	1.001121	1.007455		
04-11-2019	5374	7369.69	0.002986189	0.009212015	1.002986	1.009212		
05-11-2019	5356	7388.08	-0.00334946	0.002495356	0.996651	1.002495		
06-11-2019	5400	7396.65	0.008215086	0.001159977	1.008215	1.00116		
07-11-2019	5384	7406.41	-0.002962963	0.001319516	0.997037	1.00132		
08-11-2019	5250	7359.38	-0.024888559	-0.006349905	0.975111	0.99365		
11-11-2019	5184	7328.54	-0.012571429	-0.00419057	0.987429	0.995809		
12-11-2019	5208	7365.44	0.00462963	0.005035109	1.00463	1.005035		
13-11-2019	5342	7351.21	0.025729647	-0.001931996	1.02573	0.998068		
14-11-2019	5294	7292.76	-0.008985399	-0.007951072	0.991015	0.992049		
15-11-2019	5250	7302.94	-0.008311296	0.001395905	0.991689	1.001396		
18-11-2019	5306	7307.7	0.010666667	0.000651792	1.010667	1.000652		
19-11-2019	5502	7323.8	0.036939314	0.002203156	1.036939	1.002203		
20-11-2019	5476	7262.49	-0.004725554	-0.008371337	0.995274	0.991629		
21-11-2019	5416	7238.55	-0.010956903	-0.00329639	0.989043	0.996704		
22-11-2019	5504	7326.81	0.016248154	0.01219305	1.016248	1.012193		
25-11-2019	5544	7396.29	0.007267442	0.009482981	1.007267	1.009483		
26-11-2019	5510	7403.14	-0.006132756	0.00092614	0.993867	1.000926		
27-11-2019	5582	7429.78	0.013067151	0.003598473	1.013067	1.003598		
28-11-2019	5530	7416.43	-0.009315657	-0.001796823	0.990684	0.998203		
29-11-2019	5516	7346.53	-0.002531646	-0.00942502	0.997468	0.990575		
02-12-2019	5488	7285.94	-0.005076142	-0.008247431	0.994924	0.991753		
03-12-2019	5402	7158.76	-0.015670554	-0.017455538	0.984329	0.982544		
04-12-2019	5434	7188.5	0.005923732	0.004154351	1.005924	1.004154		
05-12-2019	5428	7137.85	-0.001104159	-0.007045976	0.998896	0.992954		
06-12-2019	5468	7239.66	0.007369197	0.014263399	1.007369	1.014263		
09-12-2019	5424	7233.9	-0.008046818	-0.000795617	0.991953	0.999204		
10-12-2019	5420	7213.76	-0.000737463	-0.002784114	0.999263	0.997216		
11-12-2019	5460	7216.25	0.007380074	0.000345174	1.00738	1.000345		
12-12-2019	5500	7273.47	0.007326007	0.007929326	1.007326	1.007929		
13-12-2019	5510	7353.44	0.001818182	0.010994752	1.001818	1.010995		
16-12-2019	5690	7519.05	0.032667877	0.022521432	1.032668	1.022521		
17-12-2019	5740	7525.28	0.008787346	0.000828562	1.008787	1.000829		
18-12-2019	5800	7540.75	0.010452962	0.002055737	1.010453	1.002056		
19-12-2019	5780	7573.82	-0.003448276	0.004385505	0.996552	1.004386		
20-12-2019	5770	7582.48	-0.001730104	0.001143412	0.99827	1.001143		
23-12-2019	5820	7623.59	0.008665511	0.005421709	1.008666	1.005422		
24-12-2019	5850	7632.24	0.005154639	0.001134636	1.005155	1.001135		
27-12-2019	5890	7644.9	0.006837607	0.001658753	1.006838	1.001659		
30-12-2019	5840	7587.05	-0.008488964	-0.007567136	0.991511	0.992433		

Valuation Methods

Valuation of Intertek under Different Approaches

- Refer the tables above for the figures

Table 2.20

1. Dividend Discount Model (DDM)

	Intertek			
	2019	2018	2017	2016
Dividend Per Share	105.8	99.1	71.3	62.4
% Change		6.8%	39%	14.26%
Total Dividend	170.8m			

Under DDM, theoretical required rate of return (r) on the company for shareholders is calculated by using Capital Asset Pricing Model (CAPM).

$$r = R_f + \beta * (R_m - R_f)$$

Risk free rate (R_f) is the GILT Bond = 1.51%

Beta / Risk (β) = 1.17 (Formula given above)

Market / FTSE 100 Return (R_m) for all the common shareholders = 12%

$$R_m = \frac{\text{Share price of FTSE as at 31}^{\text{st}} \text{ Dec, 2019} - \text{Share price of FTSE as at 31}^{\text{st}} \text{ Dec, 2016}}{\text{Share price of FTSE as at 31}^{\text{st}} \text{ Dec 2016}}$$

$$R_m = (£ 7542.44 - £ 6734.23) / £ 6734.23 = 12\%$$

$$r = 1.51\% + 1.17 * (12\% - 1.51\%) = 14\%$$

Dividend pay-outs are not constant and the company has been paying dividends since 2003. So, assuming that the company will pay dividends for 'N' number of years and the growth rate (g) will be 6.8%, which is the change from 2018 to 2019. Following is the calculation for the above Model :

$$DDM = D_1 / (r - g)$$

$$D_1 \text{ (2020 Total Dividend)} = £170.8m * 1.068 = £182.41m$$

$$\text{Value} = £ 182.41m / (0.14 - 0.068) = \text{£ 2607.80 per share price}$$

2. Earnings and Investment Based Valuation Model

Formula : $\text{Value of Intertek} = (E_1 / r) + \text{PVGO}$

Future (E_1) Earnings before interest and after tax is calculated on the basis of Growth rate (g) of 6.8%. It is assumed that under this approach growth rate will be the same as used in DDM instead of change in earnings from 2018 which is 9.8%, because of unprecedented market risk caused by Covid 19.

Theoretical rate of return (r) for discounting is also the same as used in DDM = 14%

Present Value Growth Opportunity (PVGO) formula is shown below :

$$\text{PVGO} = b * E_1 \left(\frac{\text{IRR} - r}{r} \right)$$

‘b’ is the retention rate that is 50%, given in the company 2019 financial report and ‘IRR’ is the Investment Rate of Return which is 13.6%.

- $\text{IRR} = \text{growth rate} / \text{retention rate} = 0.068 / 0.5 = 13.6 \%$

Earnings before interest and after tax in 2019 is £ 331.6m. So, 2020 Earnings (E_1) will be $\text{£ } 331.6\text{m} * 1.068 = \text{£ } 356.28\text{m}$

So now, $\text{PVGO} = \frac{0.5 * \text{£ } 356.28\text{m} (0.136 - 0.14)}{0.14} = \text{£ } - 5.09\text{m}$

Since IRR is less than the rate of return, PVGO value is negative. Hence, concluding it to be 0.

Value of the share price (VE_A) under this model is $= E_1 / r = \text{£ } 356.28\text{m} / 0.14 = \text{£}2582.72$

3. Free Cash flow Model (FCFF)

The computation of share price of the firm under this method is :

$$\frac{\text{Next year estimated FCFF to Firm (FCFF}_1\text{)}}{(\text{WACC} - g)}$$

Free Cash flow to the Firm (FCFF_0) = Earnings before interest and after tax – Capital Expenditure – Increase in Net working Capital Changes + Depreciation (2019)

- Capital expenditure is the acquisition of Property Plant and Equipment (PPE)

$\text{FCFF}_0 = \text{£}333.6\text{m} - \text{£ } 116.8 - \text{£}9.7\text{m} + \text{£}156.2\text{m} = \text{£ } 363.3\text{m}$

Growth rate (g) according to this model is the same as DDM and Earning based model i.e. 6.8%

$$\text{WACC (r)} \text{ as per this method is computed as } = \frac{\text{Equity} * \text{ROE} + \text{Debt} * R_B * (1-T_c)}{\text{Total Capital}}$$

$$\begin{aligned}\text{Total Equity} &= \text{Common Stock} + \text{Additional paid in capital} + \text{Retained earnings} \\ &= \text{£ 955.9 m}\end{aligned}$$

$$\begin{aligned}\text{ROE is the ratio of} &= \text{Net income before Interest and after Tax} / \text{Total Equity} \\ &= \text{£ 333.6m} / \text{£ 955.9m} \\ &= 34.90\%\end{aligned}$$

$$\text{Total Long-Term Debt} = \text{£ 802.2m} \quad (\text{Given : Balance Sheet})$$

$$\text{Total Capital} = \text{£1758.1m}$$

R_B is the rate of interest on the debt. So, $\text{Debt} * R_B = \text{£ 40.7m}$, given in the cash flow statement of the company and the tax rate (T_c) is given at 25.1%.

$$\text{So, WACC} = \frac{\text{Equity} * \text{ROE} + \text{Debt} * R_B * (1-T_c)}{\text{Total Capital}}$$

$$\text{WACC} = \frac{\text{£ 955.9m} * 0.3490 + \text{£ 40.7m} (1 - 0.251)}{\text{£ 1758.1m}} = 20.71\%$$

$$\begin{aligned}\text{2020 Estimated Free Cash flow to the Firm (FCFF}_1\text{)} &= \text{FCFF}_0 * g = \text{£ 363.3m} * 1.068 \\ &= \text{£ 388.0044m}\end{aligned}$$

$$\text{Value of a share by FCFF method} = \frac{\text{£ 388.0044m}}{0.2071 - 0.068} = \text{£ 2789.60}$$

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